Effect of over-actuation on neuromechanical model of rat hindlimb with biarticular muscles

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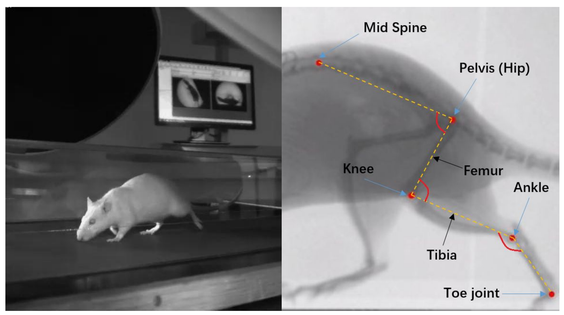
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Abstract

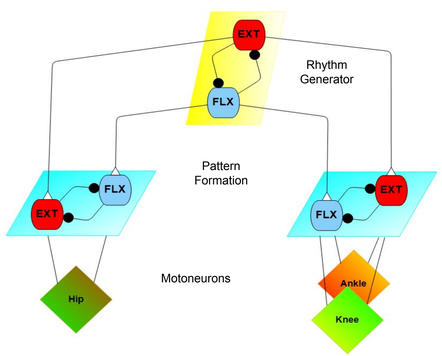
This work compares two neuromechanical models of rat hind-limb sagittal walking. One model uses antagonistic muscle pairs (extensor and flexor) actuating each joint in the sagittal plane, and the second introduces biarticular muscles into the biomechanical body. The muscle number of each limb increased from 6 to 8 in the second model, and 3 of them are biarticular muscles, BFP (biceps femoris Posterior), RF (Rectus femoris), GA (Gastrocnemii). Each model is controlled by a synthetic nervous system composed of a two-layer central pattern generator. In order to test the stability of the models, different perturbations were applied to the network. These include inhibitory tonic stimuluses to the synthetic nervous system and forces applied to the limb. Results show that the neuromechanical model with biarticular muscles exhibits more robustness than the monoarticular model when external force is applied to the body and when a stimulus is applied to the synthetic nervous system.

Background

Previously, based on studies of rat kinematics and neuron behavior from the literature, we adapted the hypothesized two-layer CPG as the controller for walking [1].

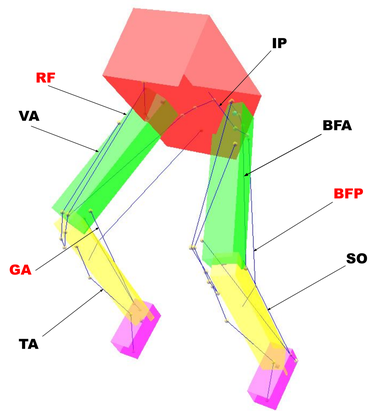


Two-layer CPG



Methods

Biomechanical Model

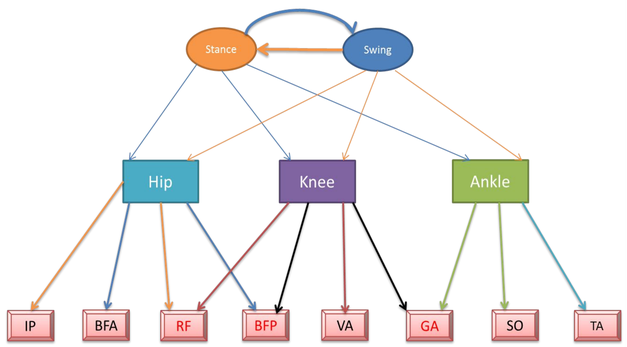


IP	Iliopsoas
BFA	Anterior biceps
BFP	Posterior biceps
RF	Rectus femoris
VA	Vastii
TA	Tibialis anterior
GA	Gastrocnemii
SO	Soleus

Rats have 38 muscles in each hind-limb. These 8 muscles were chosen because (1) parallel studies of cat hind-limb in a similar model [2]; (2) we have the EMG data for these muscles, so we can distinguish when these muscles activate during swing-stance phases.

The insertion points of these muscles come from Young's work [3] associated with the handbook "Anatomy of the rat". The muscle parameters are calculated based on Hunt's work [4].

Synthetic Nervous System

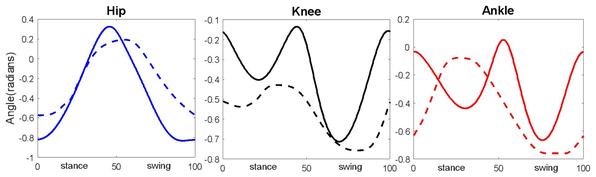


In the top layer, two neurons reciprocally inhibit each other to produce alternating flexion and extension. The 8 muscles are organized into groups and color coded in the Synthetic Nervous System (SNS).

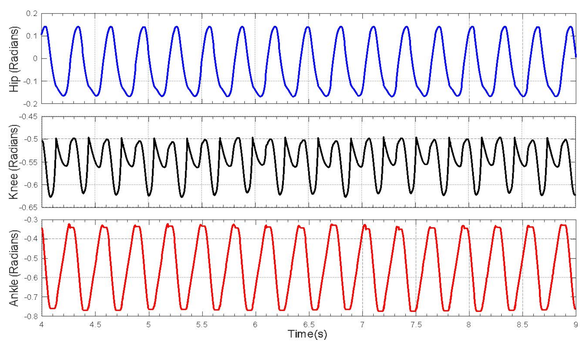
Yellow connections encourage hip flexion; Blue connections encourage hip extension; Brown connections encourage knee extension; Black connections encourage knee flexion; Green connections encourage ankle extension; Cyan connections encourage ankle flexion.

Results

Joint Motions



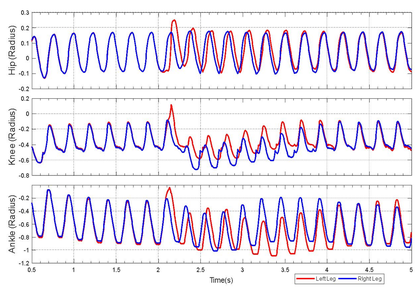
Animal data (solid lines); simulation results from monoarticular model with two- layer CPG (dashed lines);



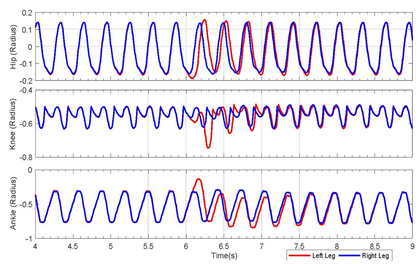
The joint motion profile using biarticular muscle shows improved walking pattern than the previous model with current parameters, the knee joint motion is strongly correlated with hip joint motion.

Stimulus Perturbation

Same inhibitory stimulus (-10 nA) applied in the pattern formation layer of the left leg for both models for duration of 0.1 s.



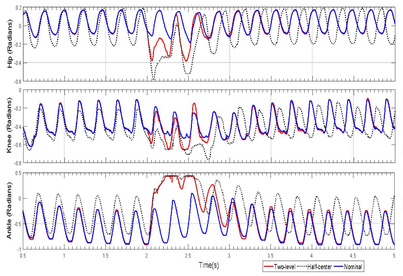
Non-resetting deletions for monoarticular model with two-layer CPG, which often restored walking rhythms after perturbation ends.



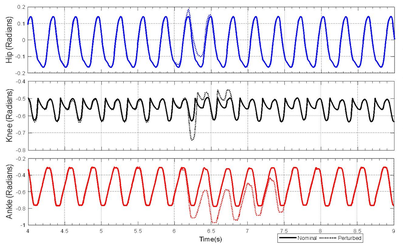
Non-resetting deletions for biarticular model with two-layer CPG, which looks to recover significantly faster and is much less perturbed, especially at the knee.

Torque Perturbation

Same Torque (0.1 Nm) applied at the Femur of left leg for both models for duration of 0.4 s.



Torque perturbation test for monoarticular model; it quickly coordinates the limb to recover stride after this external perturbation.



Torque perturbation test for biarticular model; the nominal walking pattern is restored more quickly.

Conclusions

The results of perturbation tests show that the biarticular muscle model recovers more quickly than the monoarticular muscle model after perturbation ends. When joints are linked together through muscles, rejection of a perturbation at one joint can be distributed though the whole leg.

The biarticular muscle also provides better joint motion pattern matching when compared to animal data. The biarticulate muscles can serve multiple purposes through the course of a step. While the monoarticular muscle relies on proprioceptor feedback, the biarticulate muscle mechanical linkages relate joint motion during walking.

References

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